

IMPACT LOCALIZATION BY P(VDF-TRFE) PIEZOELECTRIC NANOFIBERS SCALED UP FROM COUPON TO REAL COMPONENT

T. Ternelli^{1*}, F. Mongioi¹, B. Baldini², L. Calervo³, F. Madaro¹, F. Zonzini², R. Palazzetti³, E. Maccaferri⁴, A. Marrani⁵, C. Rossi², T. Brugo¹, A. Zucchelli¹

¹Department of Industrial Engineering, University of Bologna, Viale Risorgimento 2, 40136 Bologna, Italy

²Department of Electrical, Electronic, and Information Engineering, University of Bologna, Viale Risorgimento 2, 40136 Bologna, Italy

³Department of Mechanical Engineering, Politecnico di Milano, Via La Masa, 20156 Milano, Italy

⁴Department of Industrial Chemistry, University of Bologna, Via Piero Gobetti 85, 40129 Bologna, Italy

⁵Syensqo, Viale Lombardia 20, 20021 Bollate, Italy

*tommaso.ternelli@unibo.it

Abstract

Composite materials, renowned for their high stiffness to weight ratio, offer significant mass reductions. However, their laminar architecture makes them vulnerable to barely visible impact damage, such as delamination and matrix cracking, which can drastically degrade fatigue performance and trigger unexpected failure. Consequently, designers often rely on conservative safety factors, limiting the potential for weight savings. Structural Health Monitoring (SHM) techniques enable real-time assessment of structural integrity, permitting the design of components with lower safety factors without compromising reliability. Nevertheless, the embedding of discrete sensors, a key component of the SHM chain, introduces material discontinuities that can reduce the mechanical properties of the host laminate.

To address these challenges, this work proposes a nanostructured piezoelectric sensor fabricated from P(VDF-TrFE) nanofibers with carbon fiber prepreg electrodes, designed to minimize the mechanical impact on the host component. The sensors are co-laminated onto recycled carbon fiber prepreg components and are capable of capturing elastic waves propagating within the material, enabling the detection and localization of impact events. Impact source localization is achieved by combining the Akaike Information Criterion for Time of Arrival extraction with Gaussian Process regression, producing a probability map that identifies the most likely impact location.

To evaluate localization performance, sensors were co-laminated onto a composite battery pack casing. Preliminary tests were also conducted on a coupon plate and a module subcomponent to assess the scalability of the proposed solution. This analysis demonstrates the scalability of the methodology, showing that the proposed sensor effectively transforms complex real-world components into smart structures capable of self-monitoring.

Keywords: SHM; Impact localization; Piezoelectric nanofibers; Recycled composites; Gaussian Process Regression

1 Introduction

Composite materials are essential for lightweight and efficient modern mobility, but their widespread use generates significant waste. Among different composite recycling techniques the reuse of prepreg scraps is emerging as a viable option to offer a secondary lifecycle to otherwise wasted material [1,2]. While innovative patch architectures have resulted in significantly enhanced mechanical properties [3,4], recycled composites exhibit scattered mechanical properties due to internal discontinuities [5]. In this context, Structural Health Monitoring (SHM) enables real time assessment of the structural integrity, thus avoiding the need for excessively high safety factors in designing recycled prepreg components. In particular, given the high susceptibility of composites to impact damage, which often causes significant internal delamination without visible surface evidence, impact localization is a particularly relevant technique in the assessment of their structural integrity.

The effectiveness of the localization process depends on the precision of signal feature extraction. Many advanced methods rely on the Time of Arrival (ToA) of elastic waves triggered by the impact to compute spatial information. However, extracting an accurate ToA is particularly challenging especially in noisy environment. To address these limitations the Akaike Information Criterion (AIC) was used as it offers a robust and threshold-free method capable of precise determination of the ToA of elastic waves [6]. The differential Time of Arrival (dToA) data provided by the AIC was used to train a Gaussian Process Regression (GPR) model. Among various impact localization strategies, those based on GPR have proven highly effective in determining impact localization on composites [7]. These approaches are particularly suited for fiber reinforced materials as these probabilistic strategies generate accurate spatial likelihood maps without requiring prior knowledge of the complex wave velocity profiles typical of anisotropic materials.

However, a major limitation of traditional SHM systems is the intrusiveness of conventional sensors, such as piezoelectric (PZT) disks or Fiber Bragg Gratings (FBGs). Their integration introduces severe structural discontinuities, such as resin-rich pockets and stiffness mismatches, which can initiate delamination and compromise the host laminate's mechanical integrity [8,9]. In this context, P(VDF-TrFE) nanofiber sensors are emerging as a minimally invasive alternative to conventional sensors, thanks to their ability to detect elastic waves [10] while improving the impact resistance of the host laminates [11].

This work investigates the potential of nanostructured P(VDF-TrFE) sensors within an SHM chain that leverages Time of Arrival (ToA) extraction via the Akaike Information Criterion (AIC) and Gaussian Process Regression (GPR), assessing their scalability for impact localization through elastic wave detection. To evaluate this sensing architecture, the minimally intrusive sensors are co-laminated onto recycled carbon fiber composite components and validated through a building-block approach. The experimental campaign progresses from a baseline plate coupon to a 3D L-shaped angular profile, and ultimately to a fully assembled battery pack. This methodology explores the capability of the proposed sensors to transform passive recycled composites into smart, self-monitoring structures operating under realistic assembly conditions. Ultimately, this study demonstrates the potential of the developed piezoelectric sensor as a rapidly deployable, non-intrusive SHM solution, paving the way for the safe structural application of recycled composite materials.

2 Methods

2.1 Manufacturing of Recycled Composites Components

T700 carbon fiber epoxy prepreg (HEMT-3 CC601E, Fibertech, Marnate, Italy) was cut into 50x50 mm patches and assembled following an ordered architecture to produce equivalent laminas [4] (Figure 1a). Six equivalent laminas were stacked to shape a flat rectangular plate (290x200 mm) and four L-shaped angular profiles (540x150x120 mm). To confer sensing properties, four nanostructured piezoelectric sensors were co-laminated onto the plate and one of the angular profiles. The components were subsequently cured in an autoclave at 130°C for 1.5 hours. Once cured, the L-shaped angular profiles, including the sensorized one, were glued to two aluminum flanges to form the automotive battery pack shown in Figure 1b.

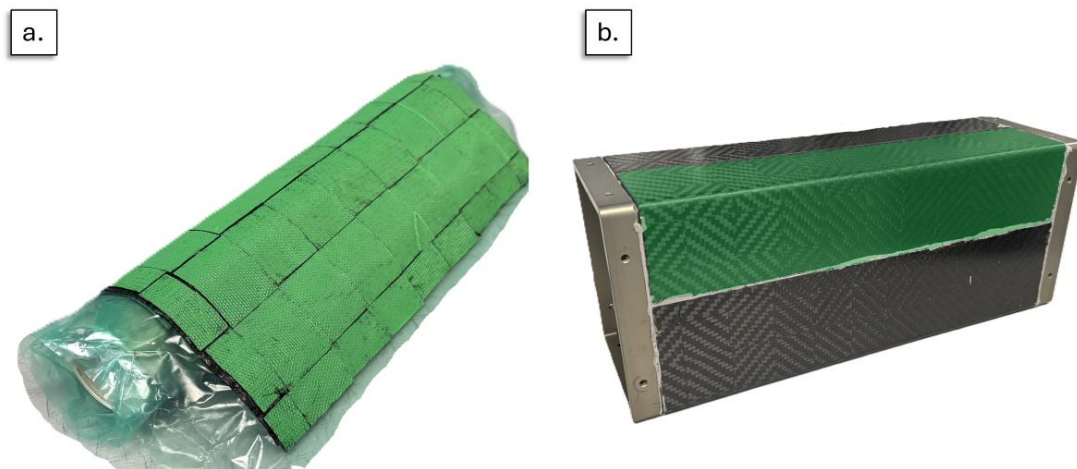


Figure 1. Recycled carbon fiber prepreg roll (a) and the assembled composite battery pack (b), with the sensorized L-shaped angular profile highlighted in green.

2.2 Electrospinning and piezoelectric sensor fabrication

The central P(VDF-TrFE) nanofiber sheets were electrospun using a needleless drum-collector machine, following the procedure detailed in [11]. The electrospinning process was conducted for 32 hours to yield a 250 μm thick piezoelectric mat. The complete sensing architecture was then manufactured by interleaving this mat with fiber reinforced prepreps, specifically, unidirectional carbon fiber (CFRP, 50 g/m², Reglass S.p.A.) and 8H Satin glass fiber (GFRP, 300 g/m², VV300S-DT121H, Delta Preg S.p.A.). As depicted in Figure 2, the resulting symmetric stacking sequence was [(GFRP 0°)/(CFRP 90°)/(CFRP 0°)/(GFRP 0°)/(CFRP 90°)/(CFRP 0°)/P(VDF-TrFE)nf]s. This specific layout leverages the CFRP plies as signal and shielding electrodes, while the GFRP layers ensure electrical insulation. Four sensors were co-cured onto each of the two selected recycled prepreg components (the flat plate and one angular profile) and subsequently poled at 100°C under a 5 kV/mm electric field to align the dipoles perpendicularly to the electrode surface.

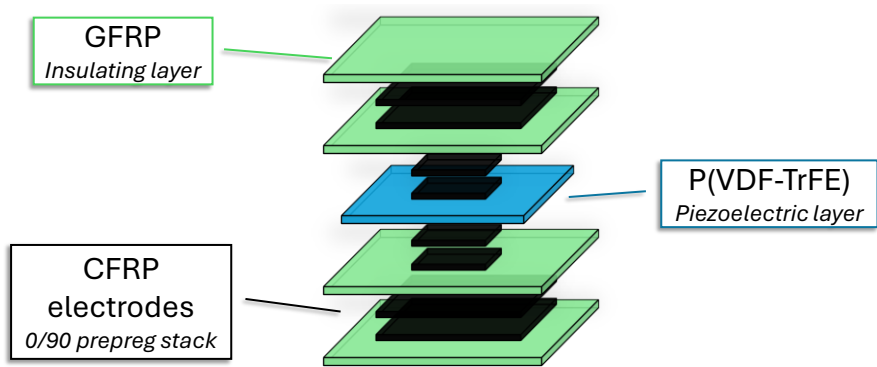


Figure 2. Stacking sequence of the piezoelectric prepreg sensor.

2.3 Impact localization

Impact localization was evaluated on three configurations: a flat plate, a standalone L-shaped angular profile, and an assembled battery pack. To minimize boundary interference, the plate and standalone angular were suspended, while the battery module was tested unconstrained to reflect operating conditions. Impacts of 12 mJ were induced by dropping a 6 g, 11.4 mm diameter steel ball from a 200 mm height across a 25 mm resolution grid. The plate utilized an 8×4 grid with supplementary points, totaling 64 impacts. The standalone angular and the assembled battery pack employed a 16×8 grid yielding 128 impacts.

During testing, sensor signals were acquired at a sampling rate of 41.7 MHz. To ensure a clean dataset prior to feature extraction, all acquired signals were processed using a 4th-order Butterworth band-pass filter. The Time of Arrival (ToA) of the elastic waves was then extracted using the Akaike Information Criterion (AIC), which adaptively identifies the signal onset without requiring manual threshold calibration. Using these ToAs, differential ToA (dToA) vectors were calculated to train a Gaussian Process Regression (GPR) model. To robustly evaluate the algorithm, the dataset was partitioned using an 80-20% train-validation split.

The GPR learns the specific mapping between known spatial coordinates and expected dToA values. During evaluation, the trained model calculates the likelihood that the measured testing dToAs originated from various points on a discretized grid, generating combined spatial probability maps. The predicted impact location is identified as the point with the highest likelihood, and localization accuracy is assessed using the Euclidean distance to the true source.

3 Results and Discussion

To assess the precision of the system, impact localization was tested across multiple structural configurations following the procedures detailed in Section 2.3. Performance was quantified utilizing the Root Mean Square Error (RMSE), with a comprehensive summary of the results provided in Table 1.

Table 1. Localization results across experimental setups

Setup	Dataset Size	RMSE (mm)
Plate	64	27
Angular	128	83
Battery	128	41

3.1 Impact localization on Coupon Plate

The flat plate achieved the highest localization accuracy with an RMSE of 27 mm. Its simple geometry enabled highly reliable ToA extraction. The 2D probability map (Figure 3) exhibits an arc-shaped distribution, reflecting the intrinsic geometric uncertainty of dToA signals, but the probability peak remains well localized near the ground truth. The simplified error map (Figure 3), displaying real impact positions as crosses and estimated positions as circles, confirms the algorithm's high precision and the absence of outliers.

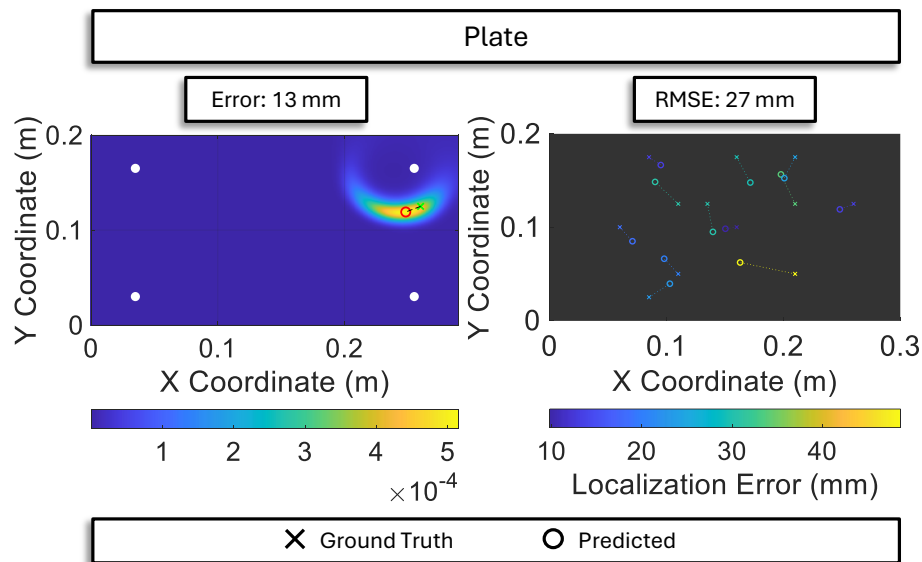


Figure 3. Impact localization for the flat plate. The 2D probability map (left) shows the spatial likelihood for a single impact, marking the ground truth (cross) and prediction (circle). The simplified error map (right) compares these coordinates across all test points.

3.2 Impact localization on L-shaped angular profile

Testing on the 3D L-shaped angular profile exhibited an overall RMSE of 83 mm. The 2D probability map (Figure 4) reveals a distribution elongated along the Y-axis, a behaviour attributed to the close proximity of the sensors along this direction, which implies a reduced dToA between them. The error map (Figure 4) illustrates that impacts located near the unconstrained boundaries display higher errors due to additional variability caused by spurious wave reflections. However, within the central area confined by the sensors, the localization performance remains highly comparable to that of the flat plate, indicating that the non-planar shape does not inherently degrade precision.

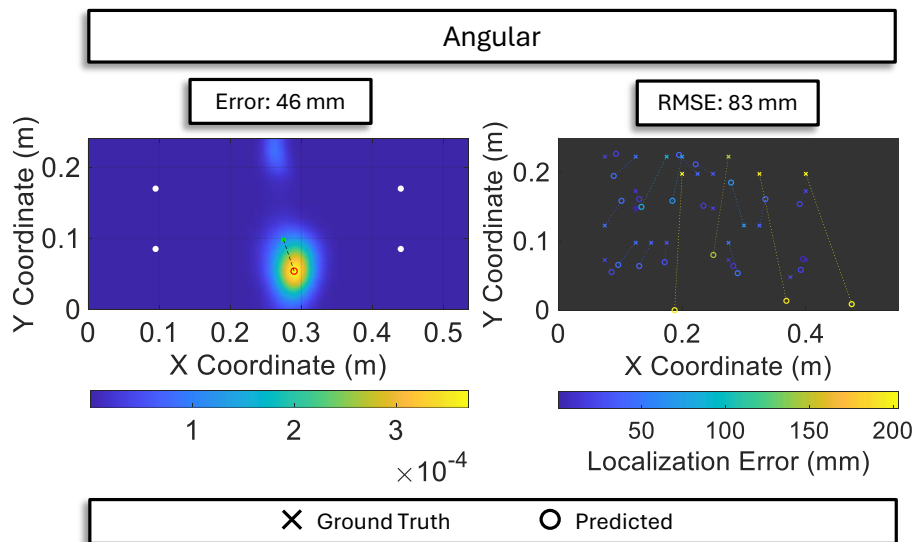


Figure 4. Impact localization for the standalone L-shaped angular profile. The 2D probability map (left) shows the spatial likelihood for a single impact, marking the ground truth (cross) and prediction (circle). The simplified error map (right) compares these coordinates across all test points.

3.3 Impact localization on Battery Pack

Once the angular profile was bonded to form the battery pack, the localization performance significantly improved, yielding an RMSE of 41 mm. The 2D probability map (Figure 5) displays a more compact and symmetric distribution, reflecting an improved confidence in the position estimate. The Y-axis elongation remains present, proving that this distribution pattern is driven by sensor placement rather than boundary conditions. The error map (Figure 5) demonstrates lower localization errors compared to the isolated angular profile. This improvement is attributed to the mitigation of spurious reflections, thanks to the material continuity ensured by the structural adhesive bonding. Ultimately, the battery pack results prove that the proposed SHM chain enables accurate impact localization within a real-world assembly context rather than an isolated laboratory coupon.

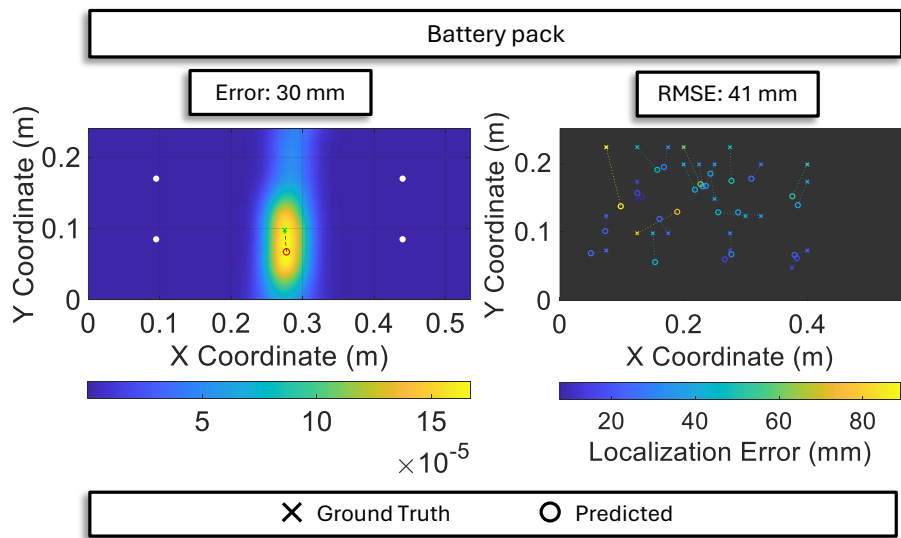


Figure 5. Impact localization for the assembled battery pack. The 2D probability map (left) shows the spatial likelihood for a single impact, marking the ground truth (cross) and prediction (circle). The simplified error map (right) compares these coordinates across all test points.

4 Conclusion

In this study, a nanostructured piezoelectric sensor incorporating P(VDF-TrFE) nanofibers was developed and tested to impart Structural Health Monitoring (SHM) capabilities to recycled carbon fiber composite components. The sensing layer was specifically designed to be minimally intrusive, aiming to preserve the mechanical integrity of the host structure. Manufacturing the sensor stack entirely from prepreg materials minimizes material discontinuities at the interface, which is expected to benefit the overall mechanical resistance of the composite. While the direct mechanical effect of the sensor integration was not explicitly evaluated in this specific work, the utilization of polymeric nanofibers has been widely demonstrated in the literature to enhance the interlaminar fracture toughness and impact resistance of composite laminates. The data-driven SHM framework, combining the Akaike Information Criterion (AIC) for time-of-arrival extraction and Gaussian Process Regression (GPR) for spatial probability mapping, proved highly robust and scalable across a building-block testing approach. The highest localization precision was achieved on a flat plate, with a Root Mean Square Error (RMSE) of 27 mm, facilitated by a simple geometry without wave propagation obstacles. When transitioning to a 3D geometry, the isolated angular profile exhibited an RMSE of 83 mm, primarily due to spurious wave reflections at the unconstrained boundaries. However, once the profile was structurally bonded to form the final battery pack assembly, the error significantly decreased to 41 mm. The material continuity provided by the structural adhesive effectively mitigated these boundary reflections, proving that the system's precision improves when applied to realistic, fully constrained structures. The successful implementation of this SHM technology represents a critical step forward for the structural application of recycled composites. By enabling continuous health monitoring, this approach effectively compensates for the inherent mechanical variability of recycled materials. Ultimately, the developed nanostructured sensor offers a reliable and highly scalable solution to transform passive composite structures into active, smart components capable of continuous impact monitoring within functional assemblies.

References

- 1 Nilakantan G, Nutt S. Reuse and upcycling of thermoset prepreg scrap: Case study with out-of-autoclave carbon fiber/epoxy prepreg. *J Compos Mater.* 2018;52:341–360.
- 2 Smith AW, Hubert P. A novel recycling framework for transforming uncured aerospace prepreg offcuts into a strand-based compression moulding compound with adjustable flow-compaction and curing behaviours. *Compos Part A Appl Sci Manuf.* 2023;175.
- 3 Palazzetti R, Calervo L, Milite A, Bettini P. Patch-Based Recycled Composites: Experimental Investigation and Modeling Techniques on Four-Point Bending and Curved Beam Traction Tests. *Polymers (Basel).* 2025;17.
- 4 Bettini P, Calervo L, Palazzetti R. Experimental study on a new generation of recycled composite laminates. *Compos Part C Open Access.* 2025;16.
- 5 Zhu TL. A reliability-based Safety Factor for Aircraft Composite Structures. *Comput Struct.* 1993;48:745–748.

Paper ID:

- 6 Pearson MR, Eaton M, Featherston C, Pullin R, Holford K. Improved acoustic emission source location during fatigue and impact events in metallic and composite structures. *Struct Health Monit.* 2017;16:382–399.
- 7 Jones MR, Rogers TJ, Worden K, Cross EJ. A Bayesian methodology for localising acoustic emission sources in complex structures. *Mech Syst Signal Process.* 2022;163.
- 8 Habib M, Lantgios I, Hornbostel K. A review of ceramic, polymer and composite piezoelectric materials. *J Phys D Appl Phys.* 2022;55.
- 9 Mastromatteo L, Gaverina L, Roche JM, Eiras JN, Irisarri FX, Lavelle F. Performance and diagnostic of guided wave PZT transducers under reusable launch vehicles thermal flight conditions. *J Intell Mater Syst Struct.* 2025;36:830–844.
- 10 Ternelli T, Mongioì F, Maccaferri E, Baldini B, Quack J, Zonzini F, et al. Impact localization in composite laminates by minimally intrusive piezoelectric nanostructured sensor. *Compos B Eng.* 2026;320.
- 11 Mongioì F, Selleri G, Maccaferri E, Fabiani D, Zucchelli A, Brugo TM. CFRP laminate with autonomous sensing and enhanced impact resistance by P(VDF-TrFE) nanofibers interleaving. *Compos B Eng.* 2025;293.